Time Series Forecasting

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Course Topics

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- ✓ What is Time Series?
- ✓ Trend, Seasonality, cyclical and random
- ✓ White Noise
- ✓ AutoRegressive Model (AR)
- ✓ Moving Average Model (MA)
- ✓ ARMA Model
- ✓ Stationarity of Time Series
- ✓ ARIMA Model
- ✓ Autocorrelation
- ✓ Akaike Information Critera (AIC)

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What is Time Series?

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- Time series is a set of observations on the values that a variable takes at different times.

- Examples: Sales trend, stock market prices, weather forecasts etc.,

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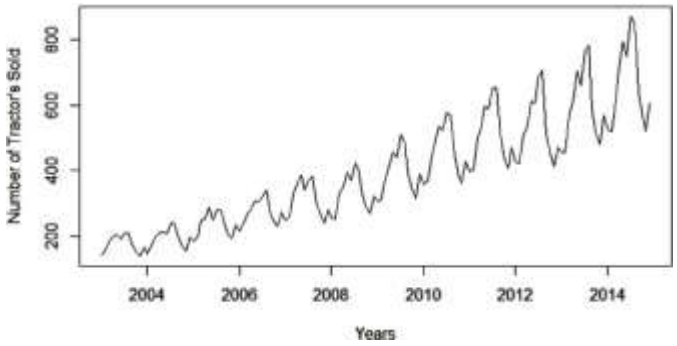
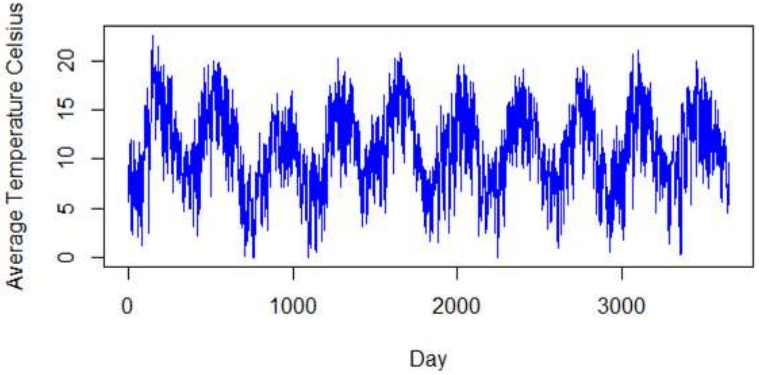
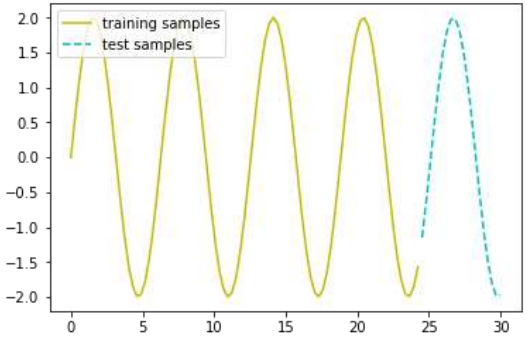
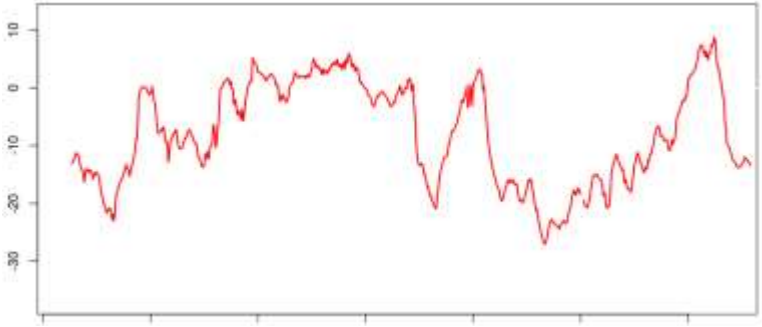
Time Series Data Patterns









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White Noise

- A series purely random in nature is called as white noise
- Mean =0, constant variance and un corellated

Average is best forecast of this series.

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AutoRegressive (AR) Model

- Y_t depends only of past values. Y_{t-1}, Y_{t-2}, Y_{t-3} etc

$$Y_t = f(Y_{t-1}, Y_{t-2}, Y_{t-3\ldots})$$

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \beta_3 Y_{t-3} \ldots$$

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Moving Average (MA) Model

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Y_t depends only on random error terms

$$Y_t = f(\varepsilon_t, \varepsilon_{t-1}, \varepsilon_{t-2}, \varepsilon_{t-3}, ..)$$

or

$$Y_t = \beta + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3} + ...$$

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ARMA Model

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Combines AR and MA

$$Y_t = \beta_0 + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \beta_3 Y_{t-3} \ldots$$

$$\varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3} + \ldots$$

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Stationarity of Time Series

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- A series is said to be “strictly stationary”, if the mean, variance and covariance is constant over period of time or time invariant
- A series which is not stationary can be made stationary after differencing.
- After differencing once, series is called as integrated of order 1 and denoted by I(1). In general I(d)

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Why Stationarity?

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- Most models assume stationarity of data. In other words, standard techniques are invalid if data is non-stationary
- Autocorrelation may result due to non-stationarity
- Autoregression results in spurious regressions

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ARIMA : AutoRegressive Integrated Moving Average

Combines all there methods.

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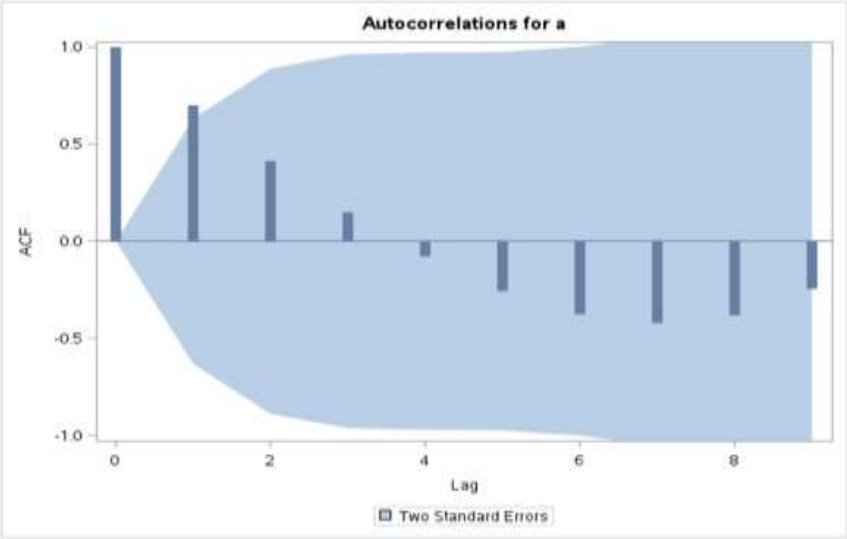
Autocorrelation

- Autocorrelation is the similarity between observations as a function of the time lag between them.



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Akaike Information Criterion (**AIC**)

- The AIC is defined by a simple equation from the sum-of-squares and number of degrees of freedom of the two models

$$\Delta AIC = N \times \ln\left(\frac{SS2}{SS1}\right) + 2\Delta DF$$

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Modeling Time Series

(attached in resource section of this module)

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Questions?

Q&A Session

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